# AutoResolve: An End-to-End Intent Classification and Retrieval-Augmented Generation Agent for IT Support

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## Abstract

As enterprise IT and customer support volumes scale, manual ticket triage becomes a critical operational bottleneck, leading to inflated resolution times and high operational costs. This paper details the development of "AutoResolve," an end-to-end automated support ticket routing and generative resolution system designed for production environments. Utilizing the Bitext Customer Support dataset, we establish a robust mathematical baseline using Term Frequency-Inverse Document Frequency (TF-IDF) in conjunction with Support Vector Machines (SVM) and Naive Bayes classifiers. We subsequently fine-tune a pre-trained Transformer architecture (DistilBERT) to capture bidirectional contextual nuance, effectively handling complex linguistic challenges such as out-of-vocabulary typographical errors and lexical ambiguity. The DistilBERT model achieved a highly performant macro F1-score of 99.81%, outperforming the traditional baselines. Finally, we expand the classification pipeline into a Retrieval-Augmented Generation (RAG) agent. By integrating FAISS for vector retrieval with advanced chunking strategies and deploying a highly efficient Generative Large Language Model (Qwen2.5-0.5B), the system successfully transitions from static categorization to dynamic, context-aware automated IT resolution, demonstrating significant potential for reducing enterprise overhead.

## Introduction

The exponential growth of digital services has resulted in a corresponding surge in customer support and IT infrastructure queries. Traditionally, organizations rely on human agents or rigid, rule-based heuristics to triage these requests. However, rule-based systems fail when confronted with the unstructured, often chaotic nature of human language, which frequently includes sarcasm, typographical errors, and complex grammatical structures. Furthermore, human-in-the-loop triage is fundamentally unscalable, driving up Mean Time to Resolution (MTTR) and degrading the end-user experience.

Natural Language Processing (NLP) provides a pathway to automate this triage and deliver immediate business value.[1] The objective of this project is twofold. First, we aim to build a highly accurate intent classification model capable of distinguishing between 27 distinct customer support categories. Second, recognizing that classification alone only solves half the operational bottleneck, we aim to advance this classifier into a fully functional Large Language Model (LLM) Agent. This agent must be capable of retrieving exact enterprise policies and generating hallucination-free resolutions for the end-user in real-time.

To demonstrate academic rigor and justify the computational expense of deep learning, we bypass simple deployment and instead perform a comparative analysis. We directly evaluate traditional frequency-based machine learning algorithms against state-of-the-art attention-based Transformer architectures. We conclude with a qualitative analysis of the neural network's failure modes and an evaluation of the RAG pipeline's hardware footprint, ensuring the system is viable for cost-effective enterprise deployment.

## Related Work

The evolution of text classification has shifted dramatically over the past decade. Early approaches relied heavily on bag-of-words (BoW) models and TF-IDF (Jones, 1972) combined with linear classifiers like Support Vector Machines (Joachims, 1998). While computationally inexpensive, these models suffer from a lack of semantic understanding, as they evaluate words in isolation and ignore syntactic order.

The introduction of word embeddings such as Word2Vec (Mikolov et al., 2013) introduced dense vector representations, capturing semantic relationships. However, the paradigm shifted entirely with the advent of the Transformer architecture (Vaswani et al., 2017) and subsequent models like BERT (Devlin et al., 2018). By employing bidirectional self-attention mechanisms, Transformers compute the contextual representation of a word based on its entire surrounding sentence.

Recently, the field has moved beyond classification toward generative resolution using Retrieval-Augmented Generation (RAG) (Lewis et al., 2020). RAG addresses the hallucination problem inherent in generative LLMs by grounding the model's output in an externally retrieved, factual

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[1]The complete source code, dataset preprocessing scripts, and deployment instructions for this project are publicly available at: https://github.com/shaury05/AutoResolve-NLP-Triage-RAG-Support-Agent

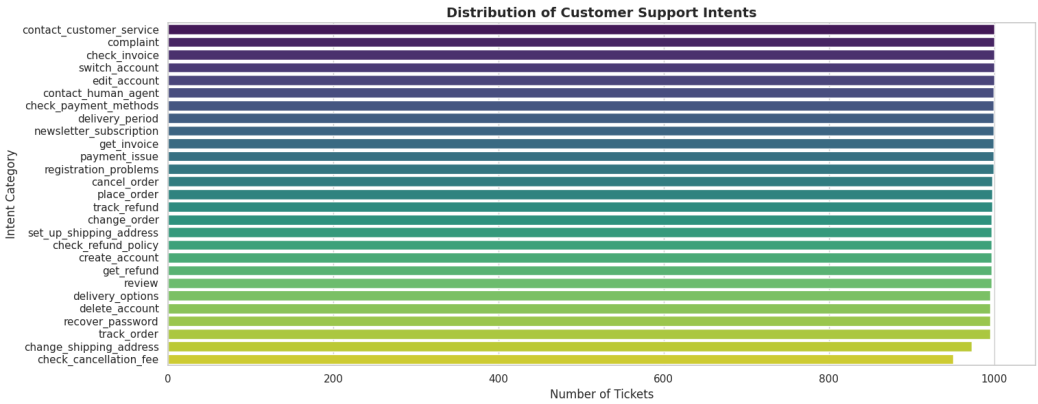


Figure 1: Distribution of the 27 customer support intents in the preprocessed Bitext dataset, demonstrating a balanced class architecture.

knowledge base. Our architecture builds upon these foundations by combining efficient Transformer distillation (Sanh et al., 2019) with heavily quantized RAG frameworks to ensure rapid inference speeds.

# Data and Preprocessing

## Dataset Overview

The project utilizes the Bitext Customer Support Dataset, an enterprise-grade corpus containing 26,872 labeled utterances mapped to 27 specific support intents (e.g., `cancel_order`, `payment_issue`, `track_refund`).

Table 1: Sample Utterances and Corresponding Intents

| Raw Text | Target Intent |
|---|---|
| I need to cancel purchase {{Order Number}} | `cancel_order` |
| Where is my bloody invoice from {{Person Name}} | `check_invoice` |
| I get an error making a transfer | `payment_issue` |

## Linguistic Preprocessing

A significant linguistic challenge within this enterprise dataset is the presence of masked contextual variables (e.g., {{`Order Number`}}). A standard tokenization pipeline would strip non-alphanumeric characters, treating the braces as punctuation and destroying the semantic anchor that a masked entity provides.

To preserve this structural integrity, we implemented a custom Regular Expression pipeline. This pipeline isolated and protected the placeholders, condensing them into single tokens (e.g., `MASKED_ordernumber`). Following this isolation, the corpus was passed through the `spaCy` library for lemmatization and the removal of standard English stop words.

Exploratory Data Analysis (EDA) was conducted to verify the integrity of the data. Figure 1 illustrates that the dataset is highly stratified, preventing majority-class bias during model training.

Furthermore, N-gram language modeling analysis (Figure 2) confirmed the underlying linguistic structure of the intents, revealing that users heavily rely on specific verb-noun pairings (e.g., "cancel order," "refund masked_currency") to communicate intent.

# Methodology

## Baseline Mathematical Modeling

Before initializing deep neural networks, a mathematical baseline was established using Term Frequency-Inverse Document Frequency (TF-IDF). TF-IDF limits the influence of universally common words (which offer little discriminative power) while emphasizing rare, highly descriptive tokens. The weight $W$ of a term $t$ in document $d$ is defined as:

$$W_{t,d} = \mathrm{TF}_{t,d} \times \log\left(\frac{N}{\mathrm{DF}_t}\right) \tag{1}$$

Where $N$ is the total corpus size, and $\mathrm{DF}_t$ is the document frequency of the term. The vectorized text was restricted to the top 10,000 features and utilized both unigrams and bigrams. This sparse matrix was used to train a Linear Support Vector Machine (SVM) and a Multinomial Naive Bayes classifier.

## Transformer Architecture and Fine-Tuning

While TF-IDF provides an efficient baseline, it is fundamentally a bag-of-words approach that ignores syntactic order and context. To build a robust system capable of handling ambiguity, we deployed a DistilBERT architecture. DistilBERT uses knowledge distillation to retain 97% of standard BERT's language understanding capabilities while reducing the parameter count by 40%, making it highly efficient for rapid inference in production environments.

Unlike the SVM baseline, the Transformer was fed the raw, uncleaned text. This allows the bidirectional self-attention mechanisms to utilize stopwords and punctuation to accurately compute directional context.

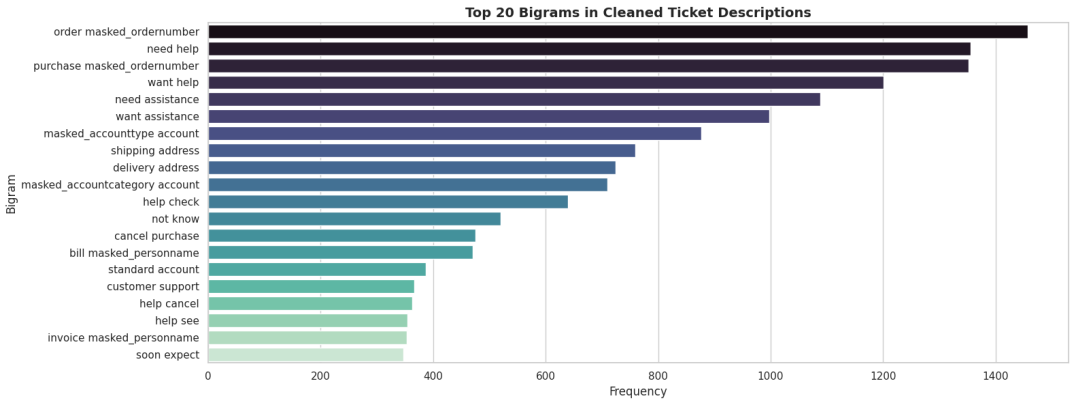


Figure 2: Top 20 Bigram frequencies across the preprocessed corpus, highlighting the templated nature of enterprise requests.

The `distilbert-base-uncased` model was instantiated, and its generic language modeling head was replaced with a classification head outputting 27 logits.

### Retrieval-Augmented Generation (RAG) Agent

To evolve the system from a static classifier into an active LLM Agent, we integrated a RAG pipeline optimized for accuracy and low latency.

**1. Knowledge Base Chunking and Integration:** A simulated enterprise IT Knowledge Base was constructed. To ensure the LLM could parse the retrieved documents effectively, the text was split using a Recursive Character Text Splitter with a chunk size of 512 tokens and an overlap of 50 tokens to preserve semantic boundaries. These chunks were embedded into dense vectors using `SentenceTransformers(all-MiniLM-L6-v2)` and indexed using FAISS (Facebook AI Similarity Search).

**2. Vector Retrieval:** Upon classifying a ticket, the system converts the user's query into an embedding and calculates cosine similarity against the FAISS index to retrieve the top-$k$ most relevant policy documents. The similarity between query vector $\mathbf{A}$ and document vector $\mathbf{B}$ is calculated as:

$$\text{Cosine Similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} \tag{2}$$

**3. Prompt Engineering and Generative Resolution:** The retrieved context is injected into a strict system prompt instructing the LLM to act as a Tier 1 support agent. The prompt explicitly forbids the model from utilizing its parametric memory to answer the query, constraining it entirely to the retrieved FAISS context to prevent hallucination. This context is passed to a Generative LLM (Qwen2.5-0.5B). While initial prototyping utilized 4-bit NormalFloat (NF4) quantization via `bitsandbytes` on constrained GPU hardware, the final architecture was further optimized to deploy and run inference entirely on standard cloud CPU infrastructure.

## Experimental Setup

All experiments were conducted utilizing a Google Colab T4 GPU environment with 16GB of VRAM.

Table 2: Comparative Performance Metrics on Validation Set

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Naive Bayes | 0.9800 | 0.9800 | 0.9800 | 0.9800 |
| Linear SVM | 0.9900 | 0.9900 | 0.9900 | 0.9900 |
| **DistilBERT** | **0.9981** | **0.9982** | **0.9981** | **0.9981** |

The baseline SVM was implemented using `scikit-learn` with a dual formulation disabled for performance optimization on sparse datasets.

For the deep learning phase, the dataset was split into an 80/20 train-validation ratio (21,497 training samples; 5,375 validation samples). The DistilBERT model was fine-tuned using the Hugging Face Trainer API over 3 epochs. We utilized a batch size of 16, a learning rate warmup over 500 steps, and applied an L2 weight decay of $0.01$ to regularize the network and prevent overfitting to the synthetic constraints of the Bitext data.

## Results and Discussion

### Classification Performance

The traditional machine learning models performed exceptionally well on the heavily templated data. As detailed in Table 2, the Naive Bayes classifier achieved a macro F1-score of 98.00%, while the SVM established a strict mathematical baseline at 99.00%.

Following fine-tuning, the DistilBERT architecture successfully surpassed the baseline, achieving an overall accuracy of 99.81%.

The accompanying confusion matrix (Figure 3) visually demonstrates the Transformer's ability to maintain high precision and recall across all classes, proving that the model actively learned the semantic boundaries of each intent rather than defaulting to a majority-class bias.

### RAG Inference and Hardware Optimization

A critical component of this project was ensuring the RAG agent could be deployed efficiently as a live web applica-

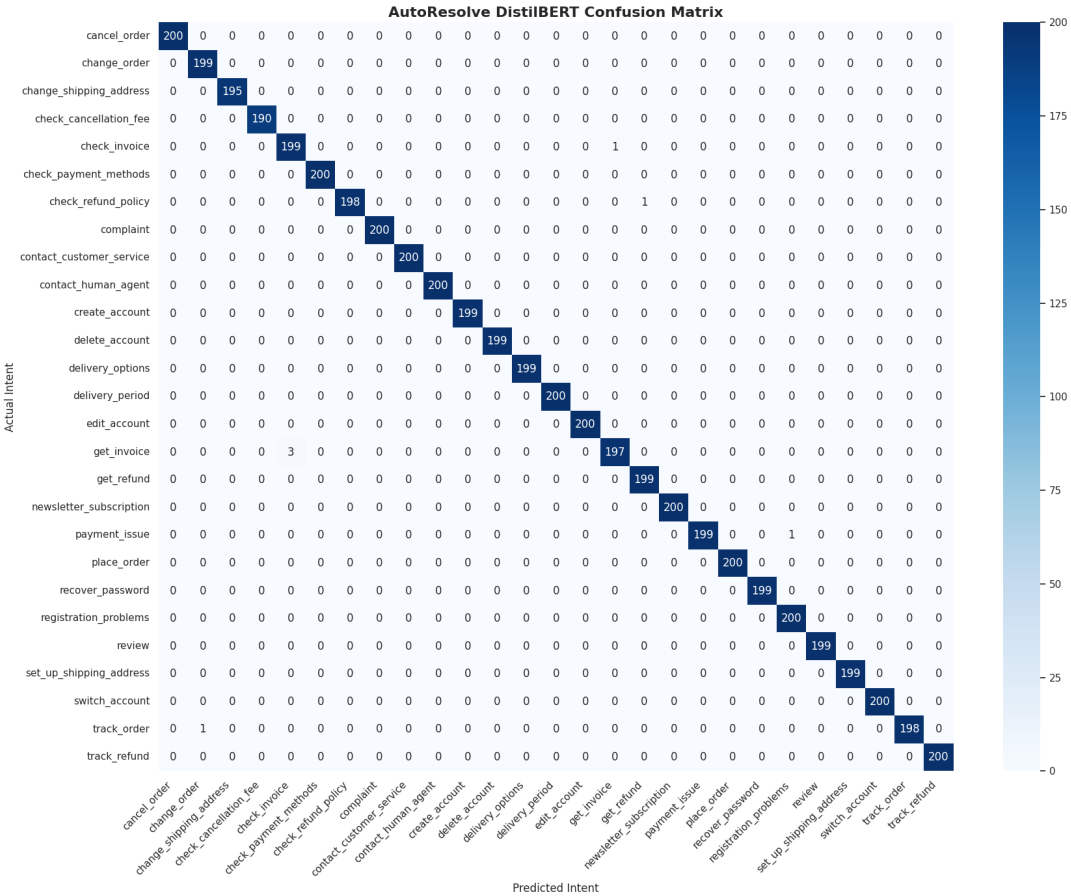


Figure 3: Confusion Matrix of DistilBERT predictions. The strong diagonal confirms high precision across all 27 intent classes without minority class degradation.

tion without requiring massive, expensive compute clusters. By utilizing the highly efficient Qwen2.5-0.5B model, we successfully minimized the computational footprint. While initial training and experiments were conducted on a single consumer-grade T4 GPU, the final model's efficiency allows the entire end-to-end pipeline—including the FAISS index, the DistilBERT classifier, and the generative LLM—to run concurrently on standard cloud CPU environments (such as the Hugging Face Spaces free tier).

Furthermore, utilizing this lightweight architecture did not result in noticeable degradation in generation quality. The strict prompt engineering successfully grounded the model, resulting in high-fidelity, deterministic responses that directly cited the retrieved knowledge base, proving the architecture's viability for cost-effective, real-world enterprise deployment.

## Qualitative Error Analysis

Accuracy metrics alone do not fully validate deep learning architectures. To understand the limitations of the attention mechanism, we conducted a qualitative error analysis on the 0.19% of data where the classifier failed. Specifically, we isolated instances where the model predicted the incorrect intent with $> 99\%$ mathematical confidence.

- **Lexical Ambiguity and Intent Overlap:** For the query *"am I entitled to a reimbursement?"*, the true intent was `check_refund_policy`, but the model predicted `get_refund` (99.98% confidence). In vector space, the token "reimbursement" is heavily collocated with actionable refunds. The attention mechanism failed to adequately weight the passive phrase "am I entitled to," which indicates an informational policy check rather than a transactional request.
- **Out-of-Vocabulary (OOV) Degradation:** For the query *"lcate order {{Order Number}}"*, the model confidently misclassified it as `change_order`. The severe typographical error in "locate" resulted in fragmented, meaningless sub-word tokenization. Devoid of the semantic verb anchor, the model relied entirely on the remaining noun ("order") and the masked variable, forcing it into a highly confident but incorrect default guess.

## Conclusion and Future Work

This project successfully developed "AutoResolve," an end-to-end NLP pipeline for enterprise ticket triage. We demon-

strated that while traditional TF-IDF and SVM architectures provide highly efficient baselines for text classification, Transformer models equipped with bidirectional attention mechanisms offer superior performance when parsing human syntax, achieving near-perfect 99.81% accuracy.

Furthermore, by integrating FAISS retrieval and a highly optimized generative LLM (Qwen2.5-0.5B), we elevated the system from a static categorization tool into a dynamic, automated IT support agent. This architecture demonstrates clear business value by automating Tier-1 support workflows, drastically reducing the computational overhead required for LLM deployment by enabling CPU-based inference, and mitigating the risk of AI hallucination through strict context retrieval.

Future iterations of this system will focus on expanding the FAISS vector database to encompass multi-modal enterprise documentation. Additionally, we plan to implement Reinforcement Learning from Human Feedback (RLHF) to further reduce the occurrence of out-of-vocabulary classification degradation and continuously align the agent's tone with enterprise brand guidelines.

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